

Predicting Hospital Readmissions

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June 24, 2026

Abstract

Hospital readmission prediction has been widely studied as readmissions can indicate increased patient risk and add extra pressure to healthcare systems. This project focuses on 30-day hospital readmission among patients with diabetes, using an existing healthcare data science study on HbA1c measurement and readmission as the starting point. It will first reproduce the key parts of the original study, including data understanding, preprocessing, the definition of 30-day readmission, and the logistic regression-based analysis. It will then extend the original explanatory analysis into a modern predictive modelling workflow. This will involve preparing the dataset for machine learning, training selected models for readmission prediction, and evaluating their performance using suitable metrics such as AUC, precision, recall and F1-score. Where possible, the project will also explore explainability methods, such as feature importance or SHAP, to understand which variables contribute most to the predictions. Subgroup performance and ethical limitations will also be considered if the available data allows this. The ideal outcome of the project would be to show that a modern machine learning approach can improve 30-day readmission prediction compared with the original statistical baseline, while still producing results that are understandable and clinically meaningful. In the best case, the project will identify useful factors associated with diabetes readmission risk and demonstrate how predictive modelling could support safer discharge planning and healthcare decision-making.

Ethics statement: This project fits within the scope of the blanket ethics application, as reviewed by my supervisor Dailia O’Grady.

I have completed the ethics test on Blackboard. My score is 12/12.

1 Project Plan

1.1 Background and Motivation

Hospital readmission is an important healthcare problem as it may indicate that a patient remains at risk after discharge or that further support is needed. Predicting 30-day readmission could help hospitals identify higher-risk patients and support safer discharge planning. This project focuses on patients with diabetes, as their readmission risk may be influenced by multiple clinical and treatment-related factors, including blood glucose control, medication changes, comorbidities, previous hospital use and demographic background.

The starting point for this project is the study by Strack et al., which analysed around 70,000 inpatient diabetes encounters and investigated the relationship between HbA1c measurement and hospital readmission rates [1]. The paper used multivariable logistic regression and controlled for factors such as age, race, admission source, discharge disposition, primary diagnosis, medical specialty and time in hospital. Its findings provide a useful baseline for this project because they connect diabetes-related clinical information with hospital readmission risk using real-world healthcare data.

1.2 Research Gap and Project Direction

Although the study by Strack et al. provides an important baseline and examines whether HbA1c measurement is associated with readmission, it does not mainly aim to build and compare modern machine learning models for individual 30-day readmission prediction. More recent research suggests that readmission risk can be influenced by a wider combination of factors, such as previous hospital use, length of stay, comorbidities and follow-up care [2].

Recent machine learning studies also show that electronic health record data can be used for readmission prediction in different ways. For example, previous work has compared logistic regression with more complex machine learning models, explored longitudinal EHR modelling, and combined manually defined clinical features with machine-learned features [3, 4, 5]. However, healthcare prediction also faces challenges such as data imbalance, data complexity, interpretability and practical implementation issues [6, 7]. Therefore, this project will reproduce the original diabetes readmission study first, then extend it into a modern predictive modelling workflow while considering model performance, interpretability and limitations.

1.3 Project Aim

The overall aim of this project is to reproduce and extend an existing diabetes hospital readmission study. More specifically, the project aims to:

- Reproduce the key data preparation and logistic regression-based analysis from the original HbA1c readmission study;

- Extend the original explanatory analysis into a predictive modelling pipeline for estimating 30-day readmission risk among patients with diabetes;
- Evaluate whether modern machine learning and explainability methods can provide useful improvement or insight compared with the original statistical approach.

1.4 Objectives

To achieve this aim, the project will:

- Review the original Strack et al. study and related literature on diabetes readmission, hospital readmission prediction and EHR-based machine learning;
- Understand the dataset structure, variable definitions, missing values and outcome definition used in the original study;
- Reproduce the main preprocessing steps, including handling repeated patient encounters, HbA1c categories, medication change indicators and the 30-day readmission outcome;
- Recreate a logistic regression-based baseline model and compare the reproduced findings with the published study where possible;
- Develop and evaluate selected machine learning models for 30-day readmission prediction, using suitable metrics such as AUC, precision, recall and F1-score;
- Use explainability methods, such as feature importance, permutation importance or SHAP, to identify which variables contribute most to model predictions;
- Consider subgroup performance and ethical limitations where possible, especially if model performance differs across demographic groups.

1.5 Expected Outcome

The expected outcome of this project is a reproducible data science workflow for diabetes-related hospital readmission prediction. The project hopes to discover whether modern machine learning methods can improve 30-day readmission prediction compared with the original logistic regression-based approach, and which clinical, demographic and treatment-related variables appear most useful for this task. The project will also consider whether the resulting models are interpretable enough to support practical healthcare decision-making, and what limitations remain when using real-world clinical data for predictive modelling.

The proposed project schedule is shown in Figure 1, and the main risks and mitigation strategies are summarised in Table 1.

References

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A Project Timeline

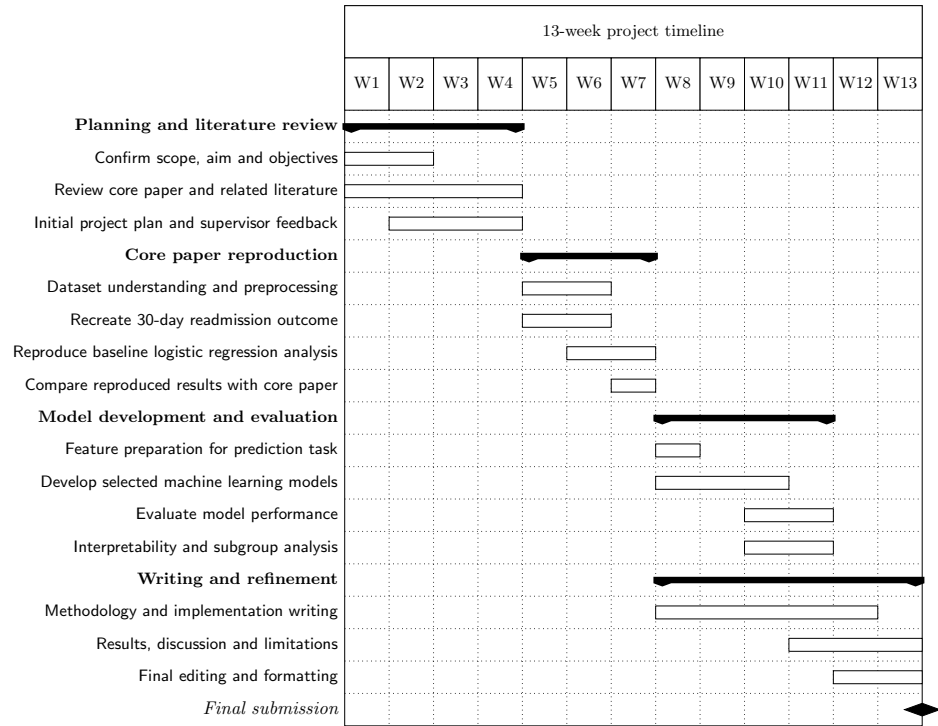


Figure 1: Gantt chart of the proposed 13-week project timeline

B Risk Assessment

Table 1: Risks and Mitigations

Risk	Likelihood	Severity	Mitigation
Difficulty reproducing the original study exactly due to unclear preprocessing details or differences in dataset handling.	Medium	High	Carefully document all assumptions and preprocessing decisions. Reproduce the original study as closely as possible, and clearly explain any differences from the published paper.
Dataset quality issues, such as missing values, noisy records, categorical complexity or class imbalance in the 30-day readmission outcome.	High	High	Perform early exploratory data analysis, use suitable preprocessing methods, and evaluate models with appropriate metrics such as AUC, precision, recall and F1-score instead of relying only on accuracy.
Machine learning models may not significantly improve over the logistic regression baseline.	Medium	Medium	Treat this as a valid project outcome. Compare models fairly, analyse possible reasons for limited improvement, and focus on what the results show about the dataset and prediction task.
Scope becomes too large due to trying too many models, explainability methods or fairness analyses.	Medium	High	Prioritise baseline reproduction and a small number of suitable models first. Add explainability and subgroup analysis only after the core modelling pipeline is working.
Ethical risk of presenting the model as clinically reliable when it has only been tested in a limited research setting.	Medium	High	Clearly state that the model is for research and analysis only, not for direct clinical deployment. Discuss limitations, possible bias, and the need for further validation before any practical healthcare use.
Fairness analysis may be limited by available demographic variables or uneven subgroup sizes.	Medium	Medium	Check subgroup performance where possible, but avoid overclaiming conclusions if the data is insufficient. Report limitations clearly in the discussion.
Technical problems, such as loss of code, data, results or computing environment issues.	Low	High	Use regular backups, version control and cloud storage. Keep scripts reproducible and save important outputs, figures and experiment settings throughout the project.